

Experiment 02: CMOS Inverter — Schematic, Layout & Verification

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Course: Analog Electronics Lab

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Github: https://github.com/heyyojas/AnalogElectronicsLab_2025BEC0010

1. Aim

To design, implement, and verify a CMOS inverter in Cadence Virtuoso — at schematic level, layout level, and through DRC/LVS/post-layout simulation.

2. Design Specifications

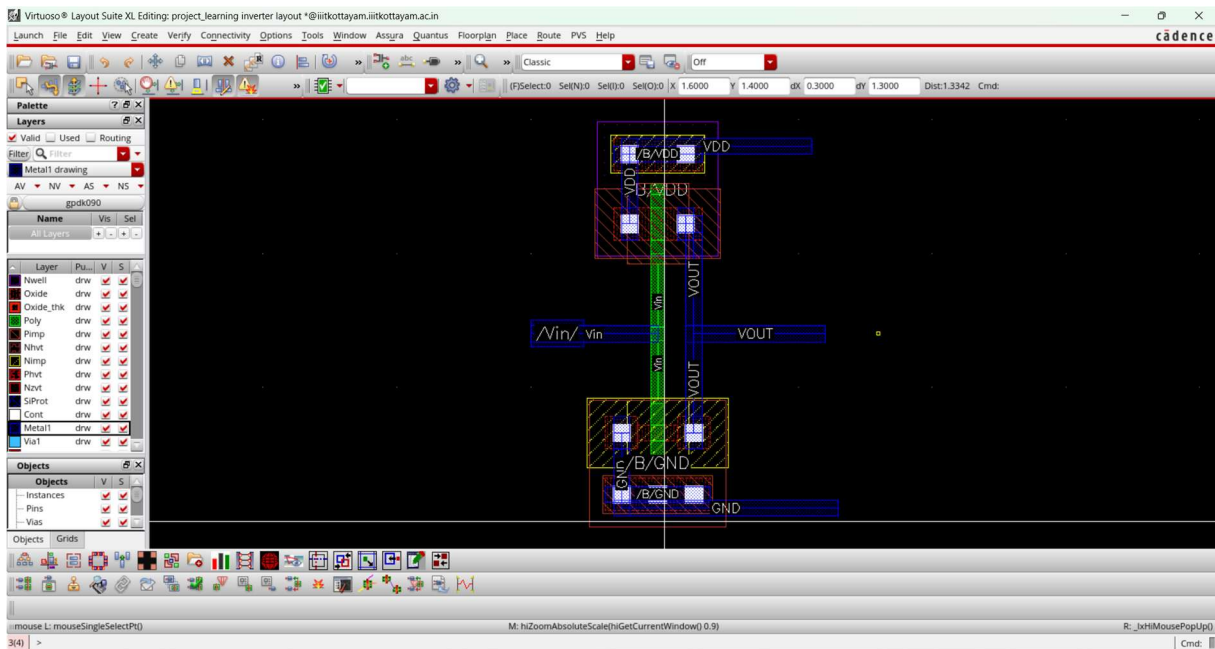
Parameter	Value
Technology	GPDK090
PMOS device	gpdk090_pmos1v
PMOS width (W)	240n
PMOS length (L)	100n
NMOS device	gpdk090_nmos1v
NMOS width (W)	120n
NMOS length (L)	100n
Supply voltage (VDD)	1.2 V

3. Circuit Description

A CMOS inverter consisting of one PMOS transistor (source at VDD, drain at VOUT) acting as the pull-up network, and one NMOS transistor (source at GND, drain at VOUT) acting as the pull-down network. Both gates are tied to the input node V_{in} , and both drains are tied together at the output node VOUT.

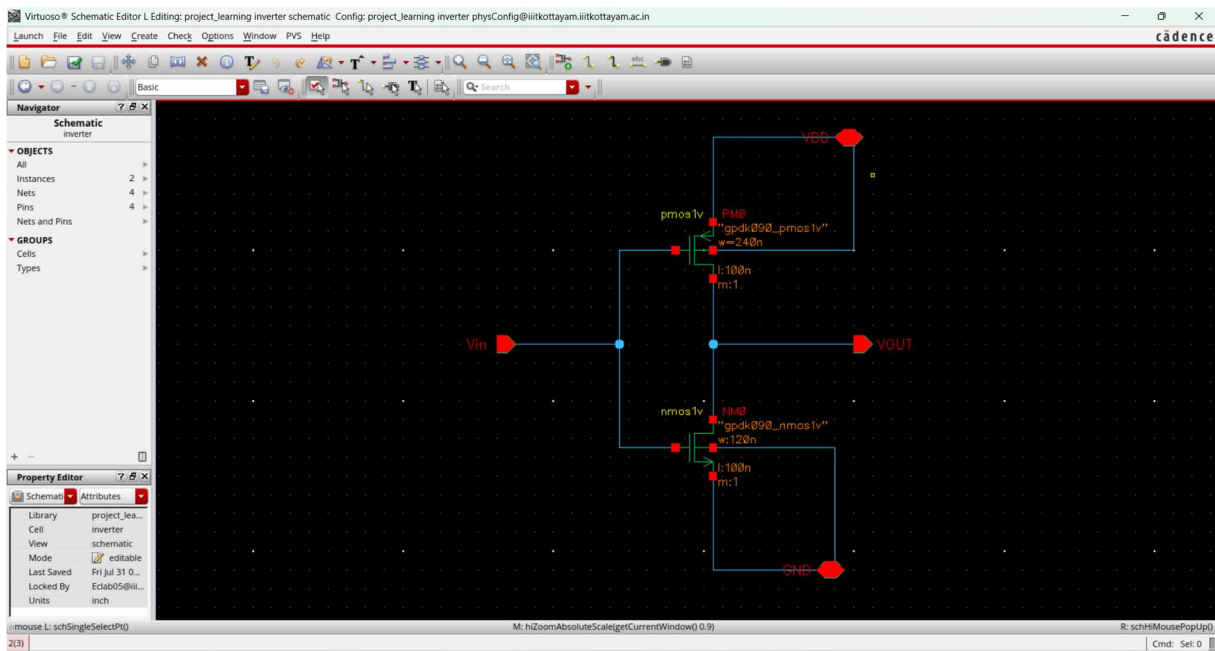
When V_{in} is high (1.2V), the NMOS conducts and pulls VOUT low (0V) while the PMOS is off. When V_{in} is low (0V), the PMOS conducts and pulls VOUT high (1.2V) while the NMOS is off — giving the expected inverting behavior.

4. Schematic



Testbench: a vdc source (V0, 1.2V) supplies VDD, a vpulse source (V1) drives Vin with V1=0, V2=1.2, rise/fall time = 1n, pulse width = 5n, period = 10n, and the gnd symbol from analogLib provides the ground reference.

5. Layout



Metal1 routing connects VDD, GND, Vin, and VOUT nets to the PMOS/NMOS transistor terminals, with well/substrate taps at the VDD and GND rails.

6. DRC (Design Rule Check)

Status: Pending — PVS tool configuration issue in the lab environment (rules file/runset not resolving). Following up with TA to identify the correct DRC invocation for this lab setup.

7. LVS (Layout vs Schematic)

Status: Pending — blocked on the same PVS toolchain issue as DRC above.

8. Simulation Procedure

Transient analysis run in ADE L (Spectre simulator), stop time = 30n, with Vin and VOUT selected as outputs. DC operating point obtained via "Save DC Operating Point" option under the DC analysis setup, annotated onto the schematic via Results → Annotate → DC Operating Points.

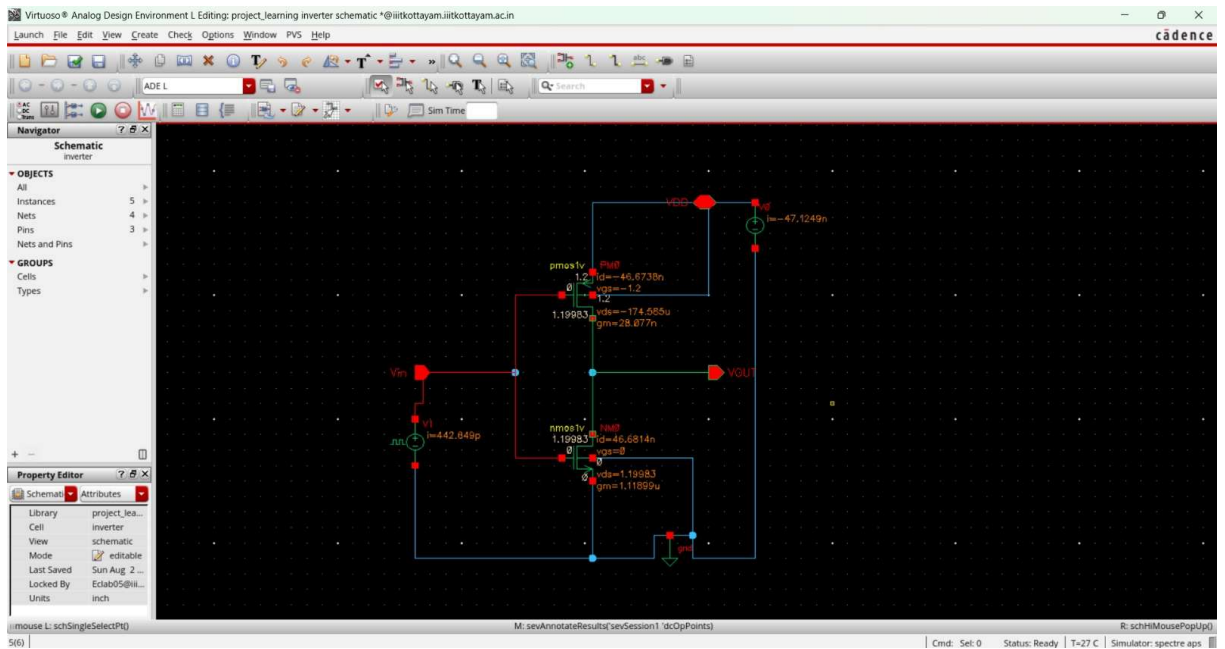
9. Results

9.1 Pre-layout Transient Simulation



VOUT cleanly swings 0V–1.2V, inverse of Vin, with sharp transition edges — confirming correct inverter switching behavior.

9.2 DC Operating Point



At $V_{in} = 0V$ (input low):

Transistor	V_{gs}	V_{ds}	I_d	g_m
PMOS (PM0)	-1.2 V	-174.585 μV (~0)	-46.6738 nA	28.077 nA
NMOS (NM0)	0 V	1.19983 V	46.6814 nA	1.11899 μA

This confirms correct operation: with V_{in} low, the PMOS is ON ($V_{gs} = -1.2V$) pulling V_{OUT} to VDD, while the NMOS is OFF ($V_{gs} = 0V$), consistent with V_{OUT} settling near 1.2V. The small nA-level currents reflect off-state leakage, as expected.

9.3 Post-layout Simulation

Status: Pending — requires LVS-clean extracted view (see Section 7).

10. Observations

The inverter's transient response confirms correct logical inversion with clean rise/fall transitions.

DC operating point values match expected MOSFET region-of-operation behavior for the given bias condition ($V_{in} = 0V$): PMOS in triode/on-state, NMOS cut-off.

DRC/LVS could not be completed due to a tool configuration issue (missing/unresolved rules file) in the lab's PVS setup — this is an environment issue, not a design issue, and is being followed up on separately.

11. Conclusion

The CMOS inverter was successfully designed and verified at the schematic level — transient simulation confirms correct inverting behavior (0V–1.2V swing, inverse of input) and DC operating point analysis confirms both transistors bias correctly for their expected on/off states. Layout was completed and DRC/LVS/post-layout verification remain pending due to a tool environment issue.